

David Israel Vázquez Leal

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Experience

AIST IRCAM (Integrated Research Center for Advanced Manufacturing)

Tokyo, Japan

Robotics engineer

September 2025 - Present

- Integrated Jetson and NUC compute stack with LiDAR, stereo cameras, and UR5e for ROS2-based mobile manipulation.
- Designed and implemented an autonomous manipulation pipeline (PCL perception, FSM task control, custom ROS2 interfaces) for robust object picking.
- Deployed Nav2 and MoveIt2 to achieve end-to-end autonomous navigation and grasping on a mobile manipulator platform.
- Authored and maintained developer documentation (Sphinx and reStructuredText) and practical usage guides (Markdown), improving reproducibility.

CNRS-AIST JRL (Joint Robotics Laboratory)

Tsukuba, Japan

Technical Staff

Sep 2024 - Jan 2025

- Developed a semi-autonomous manipulation pipeline for a mobile manipulator targeting shelf-picking of unknown objects.
- Integrated a UR5e arm with Intel RealSense D435i, ZED Mini, and Jetson Orin Nano via custom ROS2 interfaces.
- Implemented MoveIt-based motion planning and grasp pose estimation using RGB-D perception in cluttered scenes.
- First author, IEEE/SICE SII 2026: "Development of a semi-autonomous manipulation pipeline for robotic shelf-picking operations".

Roborregos

Monterrey, Mexico

Robotics Team Member (Electronics & Software)

Nov 2022 - Jun 2025

- Led the electronics and navigation subteams as Project Manager.
- Competed in RoboCup@Home 2024 (Eindhoven, Netherlands), contributing to navigation and manipulation.
- Achieved **2nd place** at the Mexican Robotics Tournament (@Home) in **2024** and **2025**.
- Co-first author: "Software Toolkit for RoboCup@Home: A Modular and Hierarchical Architecture for Service Robots" (Accepted for IEEE/SICE SII 2026).

VantTec & ZF Friedrichshafen (Self Driving Vehicle development)

Monterrey, Mexico

Robotics Research Group, Electronics Developer

Apr 2023 - Mar 2024

- Developed STM32 firmware for a full-scale autonomous braking system using CAN bus communication.
 - Designed and validated a fault-tolerant STM32-based PCB with integrated CAN transceivers to control NEMA 34 stepper motors for steering and braking actuation.
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Skills

Programming: C++, Python, C, MATLAB

Robotics: ROS2, Nav2, MoveIt2, UR5e, Robotiq, Jetson, NUC, Gazebo

Perception: PCL, RealSense D435i, ZED2/ Mini, Orbbec, LiDAR

Embedded: STM32, ESP32, FPGA, FreeRTOS, CAN, I2C, SPI, UART

Tools: GitHub, Docker, Doxygen, Sphinx, Markdown, RST, Ubuntu, KiCad/Altium

Languages: Spanish (Native), English (C1)

Education

Instituto Tecnológico de Estudios Superiores de Monterrey (ITESM)

Bachelor of Engineering in Robotics and Digital Systems (96.7/100.00)

Ceneval's EGEL Excellence Performance Award

Monterrey, Mexico

August 2021 - July 2025